

7 (a) Current I pointing downwards.

A1

Comments:

Most students obtained the correct answer.

(b) The direction of the magnetic field in the tube should be in the negative z direction.

A1

Comments:

Students are required to use Fleming's Left-Hand rule for this part.

(c)

$$F = BIL$$

$$B = \frac{F}{IL} = \frac{1.6 \times 10^4}{10.0 \times 0.300}$$
$$= 5.33 \times 10^3 \text{ T}$$

B1

A1

Comments:

Some students mistook the length as 0.100 m. Others used a product of lengths for area calculations, which is erroneous.

(d)

- A current of 10A is very large, thus a large power supply is required. Power loss due to this current will also be very high.

A1

- The large magnetic field required may interfere with the electronic equipment on the ship.

A1

Comments:

This part was poorly attempted.